

Anvit Verma

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Education

Bachelor of Technology in Computer Science

Vellore Institute of Technology

Sept 2023 – May 2027

Experience

Databricks Intern

Jio Platforms Limited (JPL)

Navi Mumbai, IN

June 2026 – Present

- Architected an automated data reconciliation engine performing daily cross-system validation of **2,300+ tables** across **14 business domains**, migrating from **SAP HANA/Oracle** into **Databricks Lakehouse** – automatically flagging missing or out-of-sync tables for remediation.
- Engineered a real-time **AI/BI** job monitoring dashboard tracking **1,100+ Databricks jobs** and **760K+ monthly runs**, using **SQL** date-spine analysis and **Quartz cron** parsing to surface an **8.2%** hidden schedule deviation masked by a false **99.9%** on-time metric.
- Automated enterprise-scale data auditing – replacing manual cross-team audits and ad-hoc **SQL** querying with self-serve analytics that cut job-investigation time by an estimated **~80%** and reconciliation effort by an estimated **40+ person-hours per week**.

Research Assistant

Vellore Institute of Technology

Bhopal, IN

Sept 2025 – Present

- Analyzed findings from **80+ peer-reviewed papers** to evaluate the evolution of **neural architectures** in autonomous driving from **Kalman filter** baselines to modern **LSTM** and **GNN** models.
- Determined a **40% reduction** in trajectory prediction error across modern safety-critical systems by validating reported **neural network** benchmarks.
- Appraised **9 foundational ADAS** surveys to identify research gaps and formulate a technical roadmap for **Neuro-Symbolic** and **Bio-Inspired** certifiable **AI** systems.

Technical Assistant Intern

Official University E-Learning Portal (Vityarthi)

Bhopal, IN

March 2025 – August 2025

- Spearheaded the end-to-end technical development of the university wide **Linux** course (**CSE0002**) for **4,100+ students** in just first semester of the launch.
- Systematized **90 Bash** and **Shell** automation scripts replicating real-world system administration workflows including log analysis and process monitoring.
- Delivered **100%** of technical screen demonstrations capturing real-time execution of advanced **Linux** commands for asynchronous learning at scale.

AI Intern

Jio Platforms Limited (JPL)

Navi Mumbai, IN

May 2025 – June 2025

- Constructed autonomous **AI agents** using **LangChain** and **LLaMA 3**, implementing a **RAG pipeline** over **500+ internal documents** for precise information retrieval.
- Implemented dynamic **SQL generation** enabling structured querying across complex relational databases within the agent workflow.
- Launched a **LangGraph**-based multi-agent system that automated **90%** of complaint routing and reduced operational latency by **20%**.

Publications

Perception Challenges in Autonomous Vehicles, International Journal 2025 Publications -

Volume 1 - Issue 3, Airo International Research Journal ISSN 2320-3714

Ishaan Shrivastava, **Anvit Verma**

Mar 2025

[\[Link\]](#)

Projects

Causal Evaluation of Maharashtra EV Policy

[\[GitHub\]](#)

Python, Polars, Pandas, SynthDiD, Scipy, Statsmodels

- Engineered a memory-efficient data engineering pipeline using **Polars** and **Pandas** to ingest, aggregate, and lazily evaluate nearly **34 million** API-sourced vehicle registrations into a balanced **16-state Parquet macro-panel**.
- Pioneered a rigorous dual-specification causal analysis using **Synthetic Difference-in-Differences (SDiD)** and Two-Way Fixed Effects, identifying a statistically insignificant **+0.035 pp ATT** (Placebo SE = 1.34) and executing a national event study on FAME-II subsidy demand displacement.

GitHub Repo Analyst AI

[\[Live Demo\]](#) [\[GitHub\]](#)

LangGraph, LLaMA 3, ChromaDB, React (Vite), Framer Motion

- Architected an autonomous **LangGraph** orchestration layer with **LLaMA 3 agents** and a **ChromaDB RAG** pipeline, enabling conversational codebase analysis, repository discovery, and automated README generation.
- Engineered an interactive **React** frontend with a flawless **Windows XP-themed desktop experience**, completely automating complex **three-stage security audits** (SAST & dependencies) directly within the OS simulation.

Tastelytics – Serverless Music Review Platform

React, AWS (Lambda, DynamoDB, API Gateway, Cognito, CDK, CloudFront)

- Architected a highly scalable, serverless **music review platform** using a monolithic **AWS Lambda** container and **API Gateway**, leveraging **DynamoDB** Global Secondary Indexes (GSI) for **sub-300ms** social feed queries.
- Fortified application security by resolving **7 critical vulnerabilities** (including IDOR and self-XSS), hardened OAuth flows natively via **Amazon Cognito**, and deployed via a **zero-downtime CI/CD** pipeline using **AWS CDK**.

Skills

Languages: Python, SQL, Linux

AI & Machine Learning: LLMs, LangSmith, LangGraph, Multi-Agent Orchestration, RAG Pipelines, Transformers, Causal Machine Learning

Frameworks & Libraries: PySpark, PyTorch, TensorFlow, LangChain, LangGraph, React, Pandas, Polars, NumPy, Scikit-learn, EconML, Statsmodels, DuckDB

Cloud & Tools: Databricks, Unity Catalog, Apache Spark, Delta Lake, Apache Airflow, Kafka, SAP HANA, Oracle, AWS (Lambda, S3, DynamoDB, API Gateway, Cognito, CodePipeline), Docker, Git, CI/CD